Subject Description Form

Subject Code	COMP2322
Subject Title	Computer Networking
Credit Value	3
Level	2
Pre-requisite / Co-requisite / Exclusion	**Pre-requisite**: COMP1011/COMP1012/ENG2002
Objectives	The key objective of this subject is to acquire a foundational understanding of computer communications technologies. Emphasis will be on the link layer and above. Networking concepts will be illustrated using the TCP/IP and ATM networks.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: *Professional/academic knowledge and skills* (a) acquire a good knowledge of the computer network, its architecture and operation; (b) understand and apply the principles and practices of computer networks; (c) realise network communication skills through programming; *Attributes for all-roundedness* (d) follow trends of computer networks; and (e) build up on team work, presentation and technical writing skills.
Subject Synopsis/ Indicative Syllabus	see table below

Topic
1. Fundamentals Networking basics; layering concept; protocols; data encapsulation; OSI reference model; TCP/IP reference model; performance evaluation.
2. Data Link and MAC Sublayer Data link layer basics; framing; error detection; automatic repeat request protocols; LAN; link layer and MAC protocols.
3. Network Layer Network layer basics; connection-oriented and connectionless networks; routing/forwarding mechanisms; distance vector and link state routing algorithms; IP basics; IP addressing and subnets; address resolution protocol.
4. Transport Layer User Datagram Protocol (UDP); Transmission Control Protocol (TCP).

<table>
<tr><td></td><td>

5. Application Layer

Networking applications.

Laboratory Experiment:

Laboratory exercises on networking such as socket programming and IP-based applications.

Case Study:

Networking technologies and applications.
</td></tr>
<tr><td>Teaching/ Learning Methodology</td><td>

Teaching is mainly conducted through lectures.

Learning is supplemented by exercises in labs/tutorials.

Students are assessed through assignments, a project and an examination.
</td></tr>
<tr><td>Assessment Methods in Alignment with Intended Learning Outcomes</td><td>

<table>
<tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="5">Intended subject learning outcomes to be assessed</td></tr>
<tr><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td></tr>
<tr><td>Continuous Assessment</td><td rowspan="3">30%</td><td colspan="5"></td></tr>
<tr><td>1. Assignments</td><td>✓</td><td>✓</td><td></td><td>✓</td><td></td></tr>
<tr><td>2. Project</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr>
<tr><td>Examination</td><td>70%</td><td>✓</td><td>✓</td><td></td><td>✓</td><td></td></tr>
<tr><td>Total</td><td>100%</td><td colspan="5"></td></tr>
</table>

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

The project is used to assess all learning outcomes.

The assignments are used to assess students’ knowledge and understanding about the subject.

Finally, students are assessed by a formal examination.
</td></tr>
<tr><td>Student Study Effort Expected</td><td>

<table>
<tr><td colspan="2">Class contact:</td></tr>
<tr><td>▪ Lecture</td><td>39 Hrs.</td></tr>
<tr><td>▪ Tutorial/Lab</td><td>13 Hrs.</td></tr>
<tr><td colspan="2">Other student study effort:</td></tr>
<tr><td>▪ Self-study</td><td>53 Hrs.</td></tr>
<tr><td>Total student study effort</td><td>105 Hrs.</td></tr>
</table>
</td></tr>
</table>

Reading List and References	**Textbook:** 1. Jim F. Kurose and Keith W. Ross, "Computer Networking: A Top-Down Approach", 7th Edition, Pearson/Addison Wesley, 2016. **Reference Books:** 1. Peterson, L. and Davie, B., *Computer Networks: A Systems Approach*, 4th Edition, Morgan Kaufmann, 2007. 2. Stevens, W. R., *TCP/IP Illustrated Volume I, The Protocols*, Addison Wesley, 1994. 3. Tanenbaum, A. S., *Computer Networks*, 5th Edition, Prentice Hall, 2010. 4. Comer, D. E., *Internetworking with TCP/IP: Volume I - Principles, Protocols, and Architecture*, 5th Edition, Prentice Hall, 2006. 5. Keshav, S., A*n Engineering Approach to Computer Networking: ATM Networks, the Internet, and the Telephone Network*, Addison Wesley Longman, 1997. 6. Stallings, W., *High-speed Networks and Internets: Performance and Quality of Service*, 2nd Edition, Prentice Hall, 2002. 7. Stallings, W., *Network and Internetwork Security: Principles and Practice*, IEEE Press, 1995. 8. Stevens, W. R., *Unix Network Programming, Volume 1: The Sockets Networking API*, 3rd Edition, Addison-Wesley Professional, 2003.